

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 21

Network Slicing

شبکه‌بندی



Lecture Notes

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1 Why Network Slicing?

Traditional mobile networks deploy a single, monolithic infrastructure designed for a “one-size-fits-all” service model. However, 5G must simultaneously serve radically different requirements:

Use Case	Requirement	Challenge
Enhanced Mobile Broadband (eMBB)	High throughput (Gbps), moderate latency	Bandwidth-hungry
Ultra-Reliable Low Latency (URLLC)	Sub-1ms latency, 99.999% reliability	Deterministic guarantees
Massive IoT (MIoT)	Millions of devices/km ² , ultra-low power	Scale and efficiency
Vehicle-to-Everything (V2X)	Low latency + high reliability + mobility	Safety-critical

Network slicing solves this by allowing **one physical network** to serve all these diverse requirements simultaneously through multiple logical networks, each optimized for its specific purpose.

Key drivers:

- **Economic efficiency:** Share expensive infrastructure (spectrum, towers, fiber) across services
- **Service differentiation:** Offer tailored SLAs to different verticals (automotive, healthcare, industry)
- **Operational agility:** Deploy new services without building new physical networks
- **Multi-tenancy:** Enable MVNOs and enterprises to operate isolated virtual networks

2 Network Slicing Concept

A **network slice** is a complete, logical end-to-end network built on top of shared physical infrastructure.

2.1 End-to-End Scope

Each slice spans the entire network:

- **RAN Slice:** Radio resource allocation, scheduling policies, beamforming configuration
- **Transport Slice:** Bandwidth guarantees, routing paths, latency bounds in backhaul/midhaul
- **Core Slice:** Dedicated or shared Network Functions (SMF, UPF, PCF, etc.)

2.2 Isolation Between Slices

Three dimensions of isolation:

1. **Resource Isolation:** Dedicated spectrum, compute, storage, and bandwidth per slice
2. **Security Isolation:** Separate security contexts, key hierarchies, access controls
3. **Failure Isolation:** A fault in one slice does not propagate to another

2.3 Customization Per Slice

Each slice can be independently customized:

- **Network Functions:** Choose which NFs are deployed (e.g., URLLC may skip certain NFs)
- **Topology:** Define the placement of NFs (e.g., edge UPF for URLLC)
- **QoS Parameters:** Configure slice-specific 5QI values, AMBR, priority levels
- **Policies:** Independent charging, access control, and mobility management rules

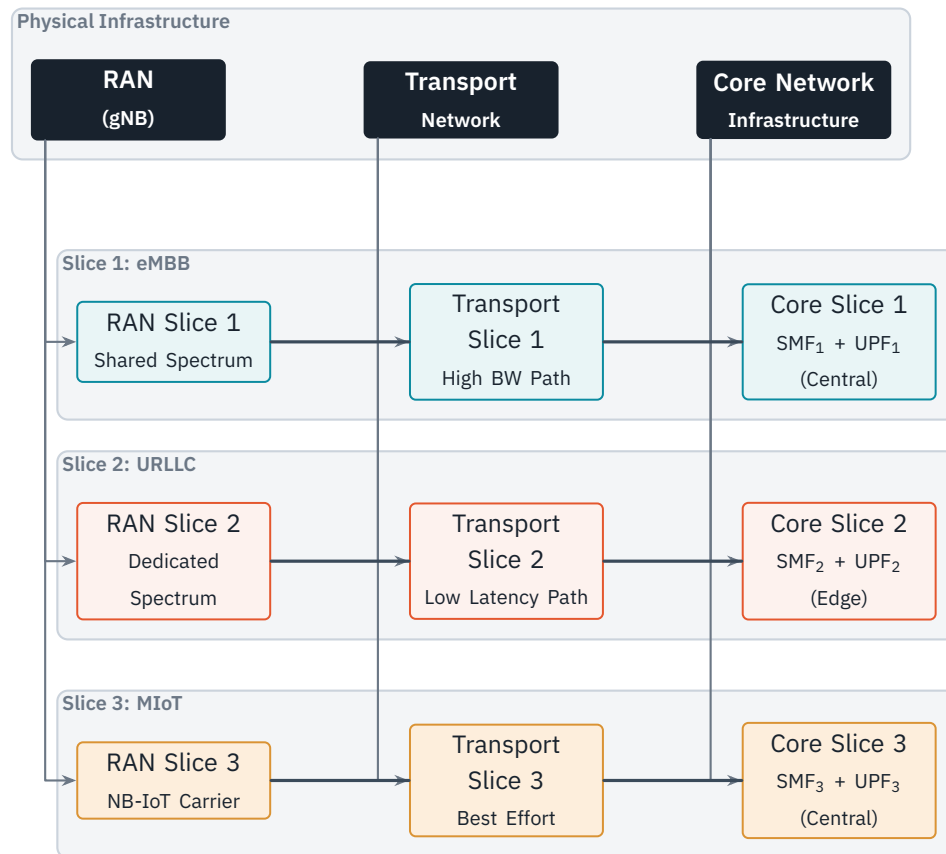


Figure 21.1 End-to-end slices across RAN, transport and core

3 Slice Identification

3.1 S-NSSAI (Single Network Slice Selection Assistance Information)

The fundamental identifier for a network slice consists of two components:

$$S-NSSAI = SST + SD$$

Field	Size	Description
SST (Slice/Service Type)	8 bits	Indicates the expected behavior of the slice (service characteristics)
SD (Slice Differentiator)	24 bits (optional)	Differentiates among slices of the same SST (e.g., different tenants)

3.2 Standardized SST Values

SST Value	Slice Type	Description
1	eMBB	Enhanced Mobile Broadband
2	URLLC	Ultra-Reliable Low Latency Communications
3	MIoT	Massive IoT
4	V2X	Vehicle-to-Everything
5–127	Reserved	For future standardized use
128–255	Operator-specific	Custom slice types

3.3 NSSAI (Network Slice Selection Assistance Information)

NSSAI = a collection (set) of S-NSSAIs. Multiple NSSAI types exist in the system:

NSSAI Type	Where Stored	Purpose
Configured NSSAI	UE (from HPLMN)	All S-NSSAIs the UE is subscribed to
Requested NSSAI	Sent by UE in Registration	S-NSSAIs the UE wants to use now
Allowed NSSAI	Returned by network	S-NSSAIs the UE is permitted to use in current PLMN
Rejected NSSAI	Returned by network	S-NSSAIs denied (with rejection cause)
Pending NSSAI	AMF internal	S-NSSAIs requiring additional authorization

3.4 Example

```

UE Subscription:
  Configured NSSAI = { S-NSSAI(SST=1), S-NSSAI(SST=1,SD=0x0000001),
                      S-NSSAI(SST=2), S-NSSAI(SST=3) }

Registration Request:
  Requested NSSAI = { S-NSSAI(SST=1), S-NSSAI(SST=2) }

Network Response:
  Allowed NSSAI = { S-NSSAI(SST=1), S-NSSAI(SST=2) }
  Rejected NSSAI = {}
  
```

4 Slice Selection (NSSF Role)

The **Network Slice Selection Function (NSSF)** is the dedicated NF responsible for determining the appropriate network slice instance(s) for a UE.

4.1 Slice Selection Procedure

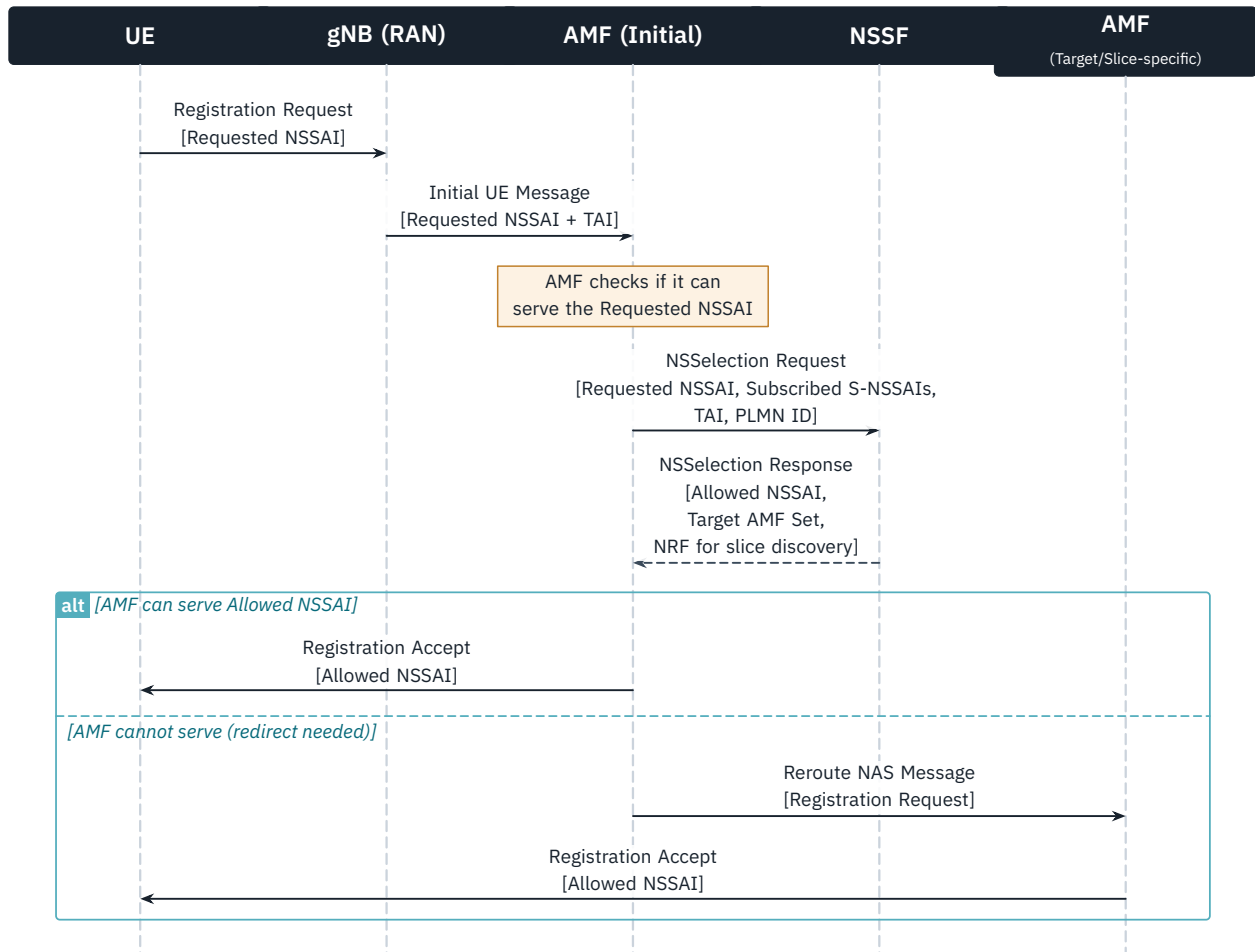


Figure 21.2 Slice selection with the NSSF and AMF re-allocation

4.2 NSSF Functions

Function	Description
Slice selection	Determine Allowed NSSAI based on subscription, PLMN policies, and availability
AMF set determination	Identify the set of AMFs capable of serving the selected slices
NRF redirection	Provide NRF endpoint for slice-specific NF discovery
Roaming support	Map between HPLMN and VPLMN S-NSSAIs

4.3 Slice Selection Inputs

The NSSF considers:

- **Requested NSSAI** from UE
- **Subscribed S-NSSAIs** from UDM (subscription data)
- **PLMN policies** (which slices are available in this area)
- **Tracking Area** (slice availability may vary by location)
- **Roaming agreements** (mapping between home and visited PLMN slices)

5 Slice Architecture

5.1 Dedicated vs. Shared NFs

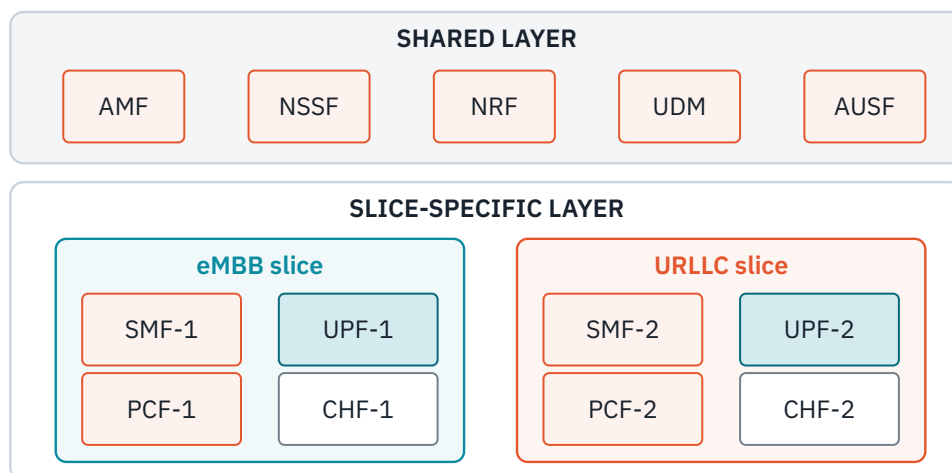


Figure 21.3 Shared network functions and slice-specific network functions

5.2 NF Deployment Model

NF	Typically Shared/Dedicated	Rationale
AMF	Shared (or per-slice set)	Handles initial access for all UEs; may have slice-specific AMF sets
NSSF	Shared	Central slice selection logic
NRF	Shared (with slice-aware discovery)	NF registry, supports slice-based queries
UDM/ AUSF	Shared	Subscription/authentication is per-UE, not per-slice
SMF	Dedicated per slice	Session management varies significantly between slice types
UPF	Dedicated per slice	Data plane customization (edge vs. central, throughput vs. latency)
PCF	Dedicated per slice	Policies differ drastically between eMBB and URLLC
CHF	Dedicated or shared	Charging models may differ per slice

5.3 Slice-Specific Customization

Aspect	eMBB Slice	URLLC Slice	MIoT Slice
UPF Placement	Central DC	Edge (MEC)	Central DC
QoS Priority	Medium (5QI=9)	Highest (5QI=80-82)	Low (5QI=79)
Redundancy	Standard	Dual connectivity, redundant paths	Minimal
Mobility	Full handover support	Conditional handover, DAPS	Rarely mobile
Charging	Volume-based	Latency SLA-based	Per-device flat rate

6 Slice Isolation

Slice isolation ensures that slices behave as independent networks despite sharing physical infrastructure.

6.1 Resource Isolation

Resource	Isolation Mechanism
Spectrum	Dedicated carriers, BWP (Bandwidth Parts), scheduling isolation
Compute	Dedicated VMs/containers, CPU pinning, NUMA allocation
Storage	Separate volumes, IOPS guarantees
Transport	VLAN/VxLAN segmentation, FlexE, dedicated wavelengths
Core NFs	Separate instances (pods/VMs) per slice

6.2 Security Isolation

- **Separate security domains:** Each slice can have independent security policies
- **Key hierarchy isolation:** Potential for separate key derivation per slice
- **Access control:** Slice-specific authentication/authorization (e.g., enterprise slice with additional EAP)
- **Traffic encryption:** Separate encryption contexts between slices
- **Audit trails:** Independent logging and compliance per slice

6.3 Failure Isolation

Failure Type	Isolation Approach
NF crash	Slice-dedicated NFs don't affect other slices
Overload	Resource quotas prevent one slice from starving others
Security breach	Compromised slice cannot access another slice's data/control
Configuration error	Slice-scoped config limits blast radius
Hardware failure	Slice-aware redundancy and recovery

6.4 Isolation Levels

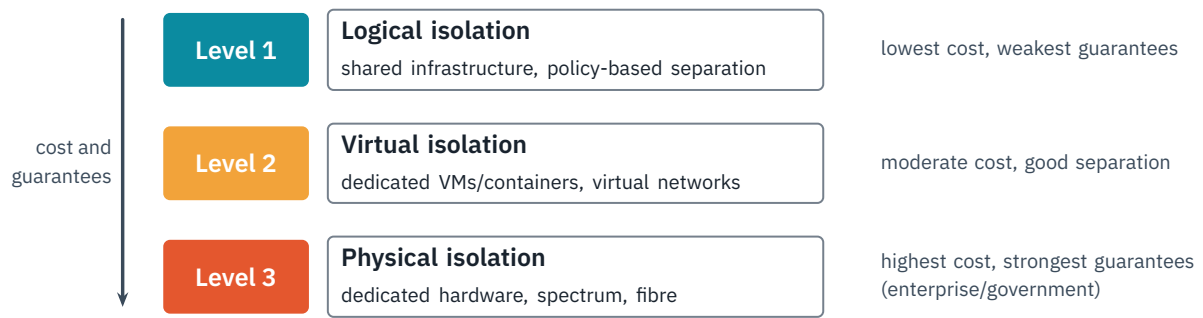


Figure 21.4 Three levels of slice isolation

7 Slice Lifecycle Management

7.1 Lifecycle Phases

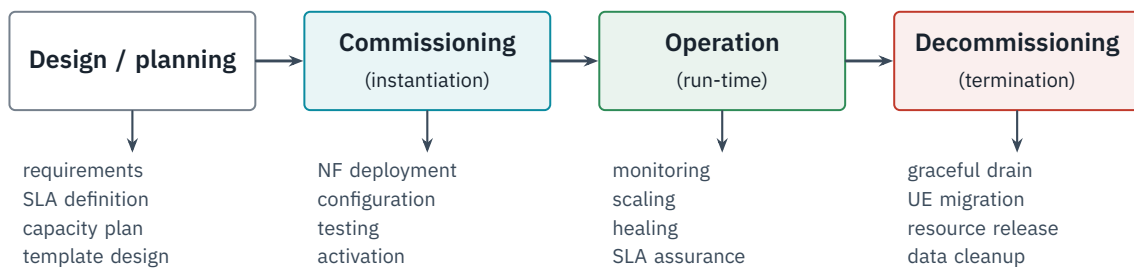


Figure 21.5 Network slice life cycle

7.2 Management Functions

Function	Role
CSMF (Communication Service Mgmt Function)	Translates customer requirements into slice requirements
NSMF (Network Slice Mgmt Function)	Manages the overall slice lifecycle
NSSMF (Network Slice Subnet Mgmt Function)	Manages individual slice subnets (RAN, Core, Transport)

Hierarchy:

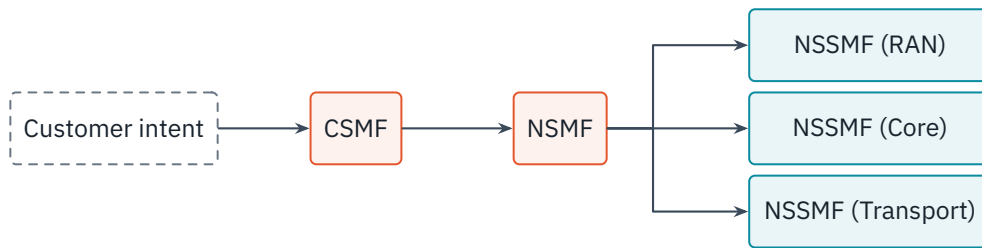


Figure 21.6 Slice-management hierarchy

7.3 Intent-Based Slice Management

Instead of specifying low-level parameters, operators express **intents**:

```

# Example Slice Intent
slice_intent:
  name: "Enterprise-Manufacturing"
  service_type: URLLC
  requirements:
    latency: "< 5ms"
    reliability: "99.999%"
    throughput: "> 100 Mbps"
    isolation: "dedicated"
    coverage: "Factory Floor Building A"
  constraints:
    max_devices: 500
    mobility: "low"
    security: "enhanced"
  
```

The management system automatically translates this intent into:

- NF deployment configurations
- Resource allocation decisions
- RAN scheduling parameters
- Transport path computation

8 Slice Use Cases

8.1 eMBB Slice

Attribute	Configuration
Goal	Maximum throughput for mobile broadband
SST	1
RAN	Shared spectrum, wide bandwidth (100 MHz+), MIMO
UPF	Central data center, high-capacity
Typical Services	Video streaming, AR/VR, web browsing
SLA	DL: 100 Mbps guaranteed, latency < 20ms

8.2 URLLC Slice

Attribute	Configuration
Goal	Deterministic ultra-low latency and high reliability
SST	2
RAN	Dedicated spectrum, mini-slots, configured grant
UPF	Edge deployment (MEC), co-located with gNB
Typical Services	Industrial automation, remote surgery, autonomous driving
SLA	Latency < 1ms, reliability 99.999%

8.3 MIoT Slice

Attribute	Configuration
Goal	Support massive number of low-power devices
SST	3
RAN	NB-IoT/LTE-M carrier, relaxed scheduling
UPF	Central, optimized for small packets
Typical Services	Smart meters, sensors, environmental monitoring
SLA	Supports 1M+ devices/km ² , battery life > 10 years

8.4 Enterprise Slice (Private Network)

Attribute	Configuration
Goal	Fully isolated private network for enterprise
SST	Operator-specific (e.g., 128)
RAN	Dedicated small cells on enterprise premises
UPF	On-premises (local breakout)
Typical Services	Campus connectivity, industrial IoT, private LAN
SLA	Full SLA customization, dedicated resources
Isolation	Level 3 (physical isolation where possible)

9 Slicing vs. QoS vs. DNN

9.1 Conceptual Comparison

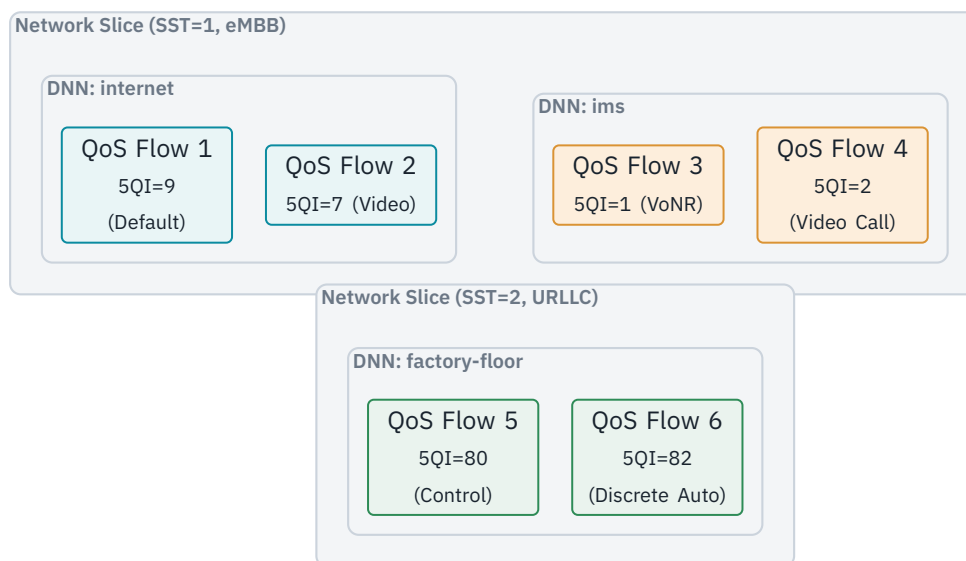


Figure 21.7 Slices contain DNNs, DNNs contain QoS flows

9.2 Detailed Comparison

Aspect	Network Slicing	QoS (5QI/QoS Flows)	DNN
Scope	End-to-end logical network	Per-flow packet treatment	Connectivity endpoint (PDU Session)
Granularity	Entire network (RAN + Core + Transport)	Individual data flows	Per PDU Session
Identifier	S-NSSAI (SST + SD)	5QI, QFI, ARP	DNN string (e.g., “internet”)
Isolation	Full (resource, security, failure)	No isolation (shared NFs)	No isolation
Customization	NF topology, placement, policies	Scheduling priority, rate, latency	Routing to specific data network
Analogy	Separate highway systems	Lane priorities on a highway	Highway exit (destination)
Who decides	Network (NSSF) at registration	PCF/SMF per session/flow	UE requests in PDU Session Establishment
Number	Few per PLMN (max 8 per UE)	Many per slice (up to 64 per session)	Multiple per slice
Relationship	Contains DNNs and QoS	Exists within a slice	Exists within a slice

9.3 Hierarchical Relationship

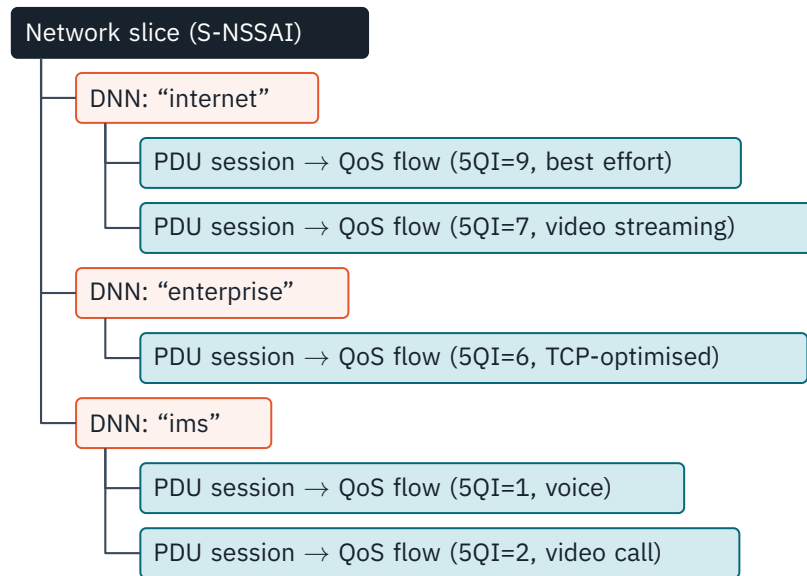


Figure 21.8 A slice contains DNNs, a DNN contains PDU sessions and QoS flows

9.4 When to Use What

Requirement	Solution
Different vertical industries on same network	Network Slicing
Prioritize voice over web traffic	QoS (different 5QI values)
Connect to different external data networks	DNN
Isolate enterprise traffic from consumer	Network Slicing
Differentiate video streaming from file download	QoS
Route to MEC application vs. central cloud	DNN (+ UPF selection)

10 RAN Slicing

10.1 Radio Resource Partitioning

RAN slicing divides radio resources among slices using different strategies:

Strategy	Description	Trade-off
Hard Partitioning	Dedicated PRBs/carriers per slice	Strong isolation, lower multiplexing gain
Soft Partitioning	Minimum guaranteed + shared pool	Good isolation + statistical multiplexing
Priority-based	Shared resources with priority scheduling	Maximum efficiency, weakest isolation
BWP-based	Assign Bandwidth Parts per slice	Clean separation at physical layer

10.2 Slice-Aware Scheduling

The gNB MAC scheduler must support multi-slice operation:

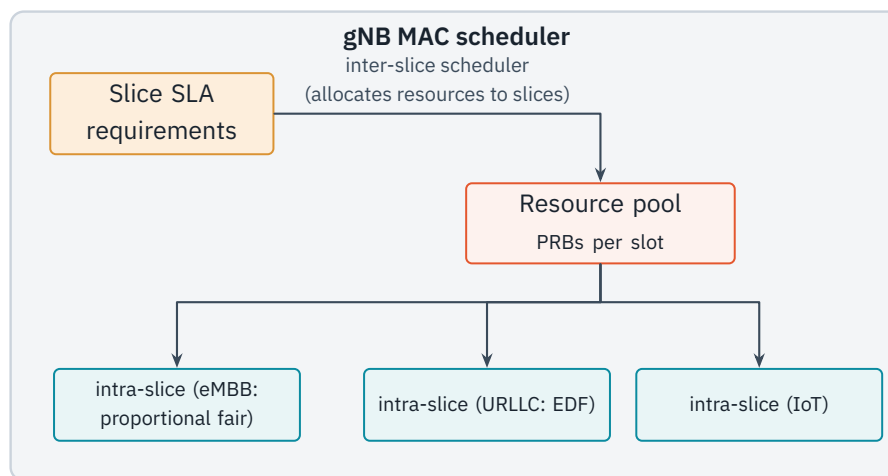


Figure 21.9 Two-level slice-aware scheduling in the gNB

Two-level scheduling:

1. **Inter-slice scheduler:** Allocates resources (PRBs, slots) to each slice based on SLA
2. **Intra-slice scheduler:** Schedules individual UEs within a slice using slice-appropriate algorithms

10.3 RAN Slice SLA Enforcement

SLA Parameter	Enforcement Mechanism
Minimum throughput	Reserved PRBs (guaranteed minimum)
Maximum latency	Priority scheduling, configured grants, mini-slots
Reliability target	HARQ configuration, repetition, diversity
Maximum UE count	Admission control per slice
Isolation level	BWP separation or hard partitioning

10.4 RAN Slicing Signaling

- **S-NSSAI in RRC:** UE provides S-NSSAI during RRC setup, gNB maps to appropriate slice
- **Slice-aware DRB mapping:** QoS flows from same slice map to common DRBs
- **RAN slice descriptors:** O-RAN defines slice-level radio resource management policies
- **NSSI (Network Slice Subnet Instance):** RAN maintains per-slice subnet instances

Summary

Concept	Key Point
Network Slice	Logical end-to-end network (RAN + Transport + Core) on shared infrastructure
S-NSSAI	SST (8-bit type) + SD (24-bit differentiator) identifies a slice
NSSF	Selects appropriate slice for UE during registration
Isolation	Resource + Security + Failure isolation between slices
Lifecycle	Design → Commission → Operate → Decommission (managed by NSMF/NSSMF)
RAN Slicing	Radio resource partitioning with two-level scheduling
Slice vs QoS vs DNN	Slice (whole network) $\supset$ DNN (connectivity) $\supset$ QoS (per-flow treatment)

Key Takeaways

1. **Network slicing is the defining feature of 5G architecture** — it transforms a single physical network into multiple purpose-built logical networks
2. **S-NSSAI = SST + SD** is the fundamental slice identifier; UE provides Requested NSSAI, network returns Allowed NSSAI
3. **NSSF is the brain of slice selection** — it determines which slices a UE can access and which AMF should serve it
4. **Isolation is multi-dimensional** — resource, security, and failure isolation must all be addressed
5. **RAN slicing requires two-level scheduling** — inter-slice allocation followed by intra-slice per-UE scheduling
6. **Slicing $\neq$ QoS** — slicing provides end-to-end logical network separation; QoS provides per-flow treatment within a slice
7. **Lifecycle management** automates slice creation through decommissioning, driven by intent-based specifications